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Numerical Simulation of Deep Multilateral Closed-Loop Well System for Geothermal Energy Extraction

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Abstract

The energy transition requires efficient alternative energy sources to replace traditional fossil fuels. Unlike other renewable energy options, geothermal energy is a reliable solution as it can be harnessed anywhere and at any time, and it is non-depletable. Geothermal energy offers a sustainable, low-carbon solution to the world's energy needs, but efficiently extracting it from deep reservoirs remains a significant challenge. The closed-loop technique has emerged in recent years as a new trend that overcomes the limitations of classical geothermal energy exploitation. This study presents a numerical simulation of a deep, multilateral, closed-loop well system designed to enhance the efficiency of geothermal energy extraction. The closed-loop configuration eliminates the need for direct fluid interaction with the reservoir, thereby addressing issues such as scaling, corrosion and depletion of the reservoir. Furthermore, the multilateral system is expected to outperform the single closed-loop well system, which has been extensively discussed in the literature to date.

The numerical model incorporates heat transfer processes within the wellbore and the surrounding geological formations, taking into account varying reservoir temperatures, thermal conductivities, and flow rates. Up to 12 multilateral pairs were simulated in this study. In the horizontal sections, water contacts the wellbore wall directly to avoid reducing the heat conduction rate from the rock to the wellbore. The proposed model was created and solved using finite element software, which handled the wellbore path efficiently to provide an acceptable CPU time. A comparison with a single closed-loop system in vertical and horizontal configurations is also presented to justify the use of a multilateral system.

Simulations demonstrate the system's thermal performance, highlighting the key parameters that influence energy output. These include the spacing and length of lateral wells, and the effect of cooling at the injection site. The results also show that multilateral configurations significantly improve heat exchange efficiency compared to single-loop systems, making them a promising option for harnessing deep geothermal energy. Such systems can also be used for electricity generation, which is an additional advantage over single closed-loop wells. Advancements in tubular insulation and new open-hole techniques without casing are expected to drive the future success of this technology.

These findings provide critical insights for optimizing system design and scaling up geothermal technology for commercial applications. Additionally, this study offers a preliminary cost estimate for a similar project initiated in mid-2023 in Germany, expected to be operational by mid-2025.

Keywords: Closed-loop, Finite element method, Geothermal, Geothermal energy, Multilateral, Multilateral closed-loop

Introduction

Since the beginning of the Industrial Revolution, fossil fuel energy based on coal, oil and gas has been used on a large scale. These fuels pollute the environment by releasing billions of tons of CO₂ and other greenhouse gases into the atmosphere every year. Climate change, caused by heat being trapped in the atmosphere, has led to serious changes and catastrophes that need to be addressed. However, due to the current industrial framework's dependence on a continuous fossil fuel supply, it is not feasible to cut production and consumption immediately. Therefore, some CO₂ removal technologies from the atmosphere, primarily known as Geological Carbon Storage (GCS), are now within the remit of the exploration and production (E&P) industry to offset the greenhouse gases produced by the combustion and production of hydrocarbons. Although this technology is seen as the primary way to reduce CO₂ emissions, the necessary reductions in greenhouse gas emissions and global temperature rise will not be achieved as long as the industrial value chain relies on fossil fuels due to the slow advancement of the technology and the legislative challenges associated with GCS. Therefore, the method of mitigating the climate crisis should consist of several decarbonization methods, as well as a mutually beneficial relationship (economic, structural and public acceptance) between these decarbonization value chains. Fig. 1 shows global direct primary energy consumption between 2000 and 2023 (Energy Institute, 2024). It can be seen that fossil fuel consumption continued to increase up to the end of 2023, while the share of renewables in the energy mix remains modest.

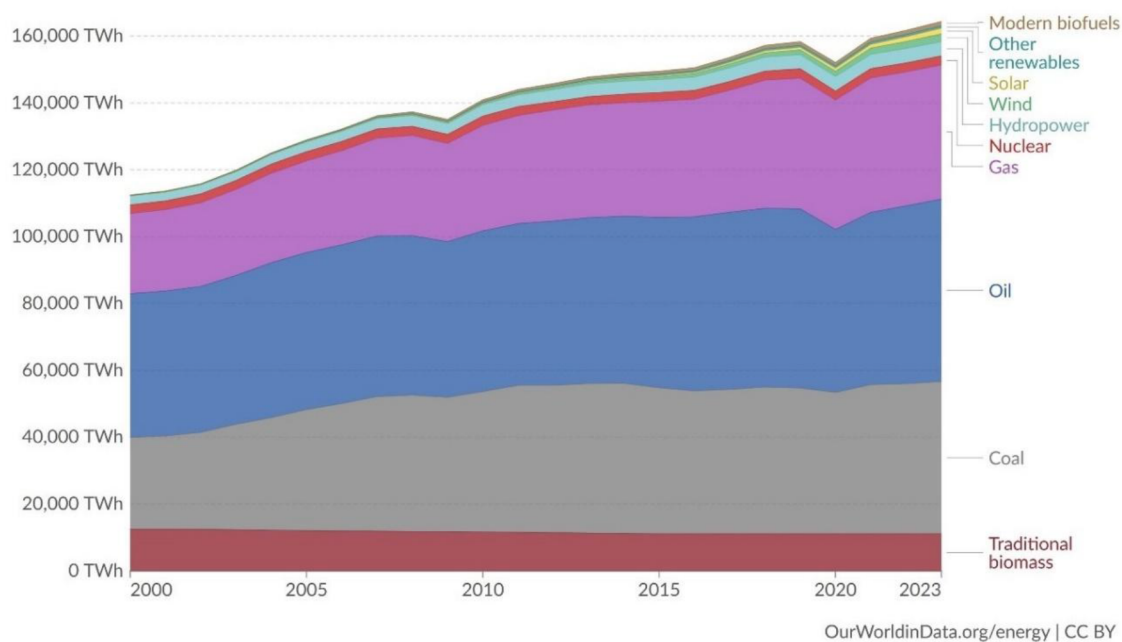


Figure 1—Global direct primary energy consumption between the years 2000-2023 (after Energy Institute, 2024). Energy consumption is measured in terawatt-hours, in terms of direct primary energy. This means that fossil fuels include the energy lost due to inefficiencies in energy production.

The development of renewable energy sources such as wind and solar has been remarkable. However, these resources suffer from intermittency, which makes them insufficiently reliable on their own. In recent

years, new technologies have been developed to store excess energy from these sources, enabling the creation of a more dependable energy system powered by renewables. Hydrogen is widely considered a promising solution to this challenge. Surplus energy can be converted into hydrogen through water electrolysis and stored underground in geological formations on a large scale (millions of tons). Later, during peak demand periods—whether daily or seasonal—the hydrogen can be retrieved and converted back into electricity or directed to other energy-consuming sectors. Nevertheless, this process remains technically challenging, economically uncertain, and requires further research to reduce overall costs.

Geothermal energy is abundant, continuously available 24/7 throughout the year, and unaffected by weather conditions. In the past, large-scale exploitation of geothermal energy was limited due to insufficient technology. However, over the past two decades—especially in recent years—new techniques have emerged that have significantly simplified the implementation of geothermal projects in various regions. It is estimated that approximately 2×10^{31} joules of geothermal energy are stored beneath Earth's crust (Möller 2020). A portion of this vast energy reserve could supply the entire planet with clean, reliable, and uninterrupted energy for heating, electricity generation, or power-to-X technologies.

Classical geothermal projects can be categorized into shallow geothermal systems and deep geothermal doublets. In this study, shallow geothermal systems are not discussed, as they fall outside the scope of this work. Traditional geothermal doublets involve drilling two wells: one for water injection and the other for the production of hot water or steam. This technology is well understood, extensively documented, and standardized within the geothermal energy sector. However, it faces several challenges—some of which are difficult to mitigate—including corrosion, mineral precipitation, injectivity issues, and induced seismicity (Nassan et al. 2024, 2025). Additionally, the success rate for drilling geothermal wells is relatively low, with Sass et al. (2016) reporting a rate of approximately 40%.

Closed-loop geothermal system

The closed-loop system (CLS) is an alternative approach to geothermal energy extraction using doublet (Liu and Taleghani 2023). The principle of CLS depends on circulating water or any other fluid within one well or two connected wells (Fig. 2). One of the main advantages is eliminating contact between the circulated fluid and the geological formations. Over the past two decades, this method has been widely studied and tested in several locations worldwide. It offers solutions to many of the challenges faced encountered by traditional geothermal systems while significantly reducing project costs (Nassan et al. 2024). Unlike open-loop systems (doublet system), closed-loop technology can be deployed in diverse locations, including areas with numerous abandoned or soon-to-be-abandoned wells, as well as wells originally drilled for other purposes (such as geological research, storage, or unsuccessful exploration). Many of these wells are situated in regions with high geothermal gradients, making them ideal candidates for closed-loop geothermal energy extraction.

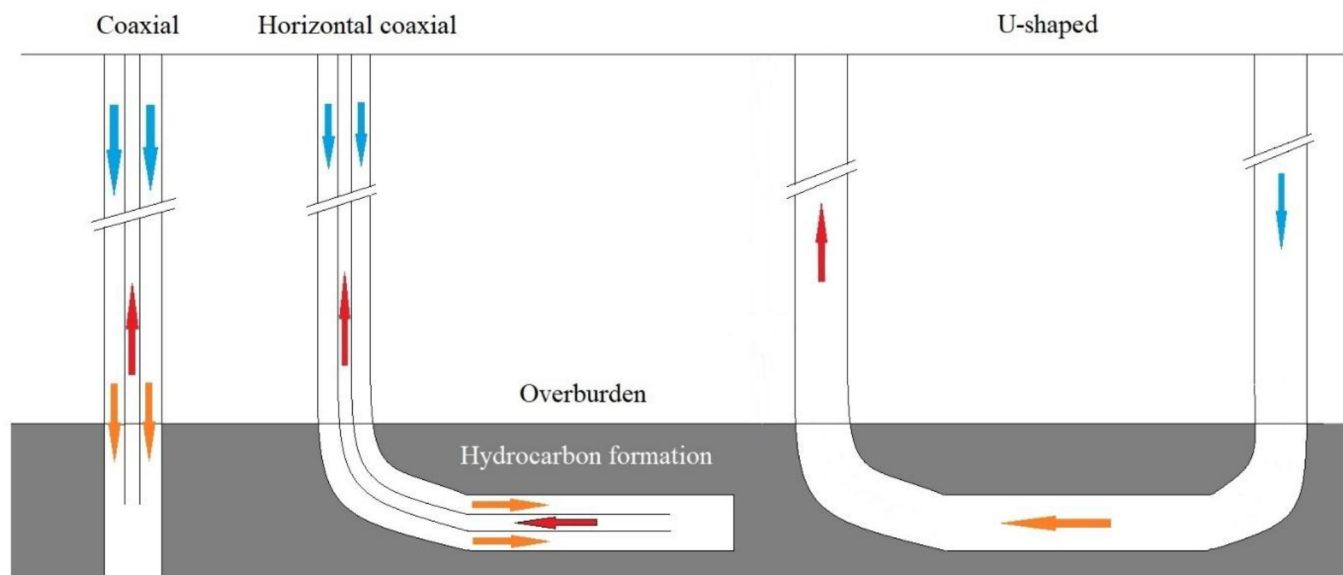


Figure 2—Main CLS configurations reported in the literature (after Nassan et al. 2024). Other, less common configurations have also been proposed.

CLS rely mainly on heat conduction between the geological formations and the wellbore as depicted in Fig. 3. As can be seen, convection can play a role in the process of heat transfer to the wellbore when a fluid exists within the pore space of a high permeable formation.

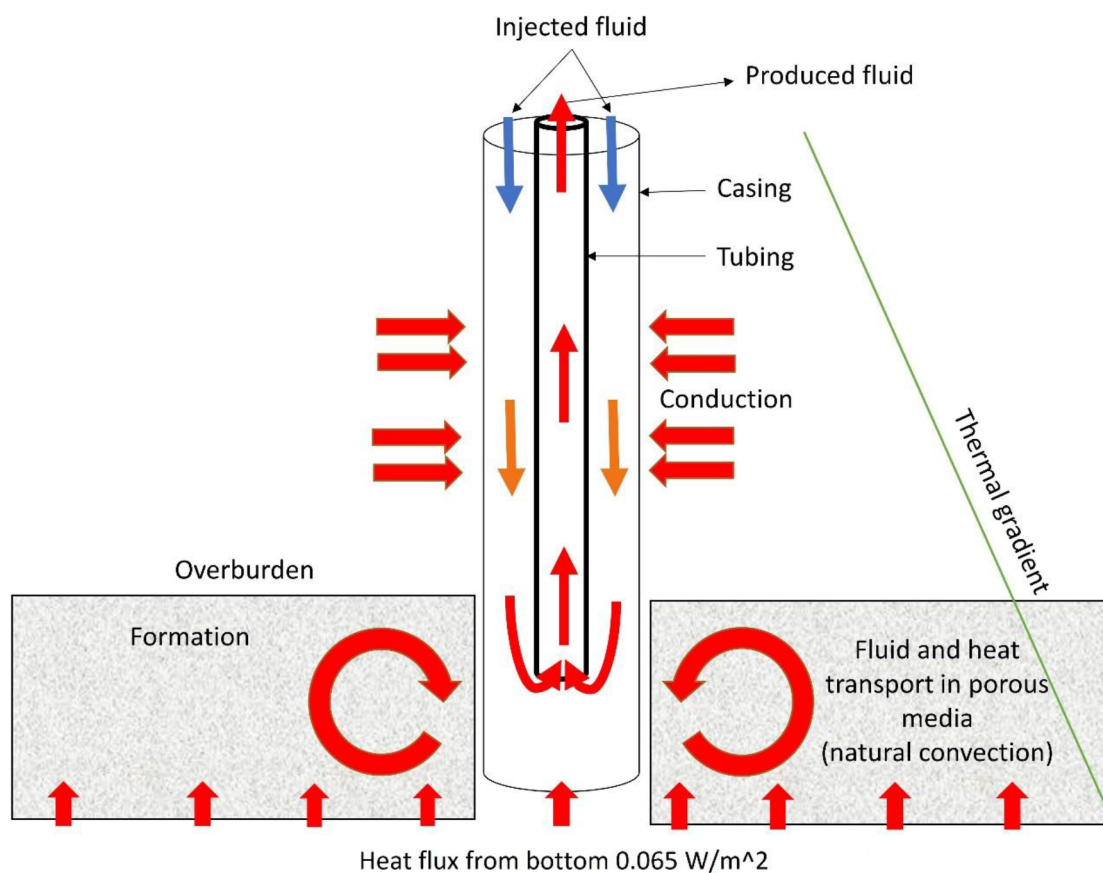


Figure 3—Mechanisms of heat transfer between the wellbore and the surrounding formations (after Nassan et al. 2024).

Multilateral CLS is a new concept that have been introduced in the last five years and increasingly being considered by the geothermal energy sector, which is moving toward this approach. Fig. 4 illustrates potential configurations of multilateral branches expected to be implemented soon. One example is the stacked Closed-Loop project in Geretsried, Germany, which aims to provide district heating and electricity to an entire region within the next three years using a similar closed-loop system configuration. The target formation is located at a depth of 4,500 meters, with a temperature of around 165 °C, and drilling began in July 2023. The first phase of this unique project is expected to be brought into operation by mid-2025 (the whole project comprises of four phases). As of Mid-May, 2025, six multilateral pairs were drilled, where each pair consists of two horizontal sections. In such systems, decision-making must balance increased project costs with higher heat extraction rates.

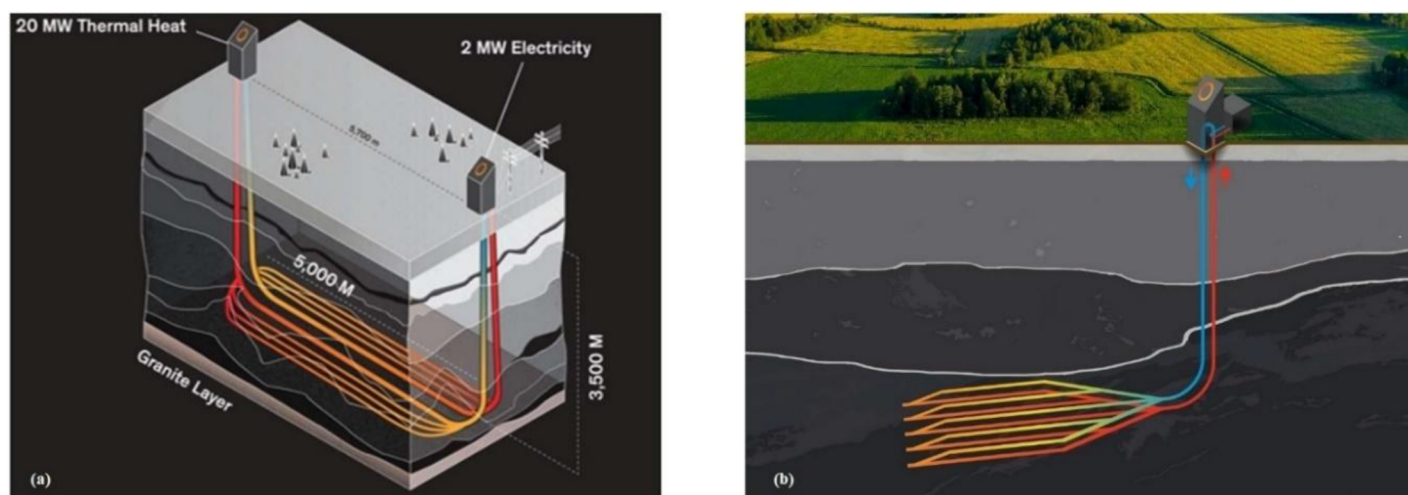


Figure 4—Original configuration of stacked closed-loop (after Nassan et al. 2024): a) Multilateral U-shaped configuration; b) Planned stacked closed-loop configuration for Geretsried-Germany geothermal project: twelve multilateral pairs are connected to two wells where one well for injection and the other for production.

What makes the stacked Closed-Loop unique compared to traditional geothermal systems is that it operates as a closed-loop system, eliminating the need for permeable rock formations or hydrothermal reservoirs. Below are some of its key technical features CLS project in Geretsried (Geretsried project 2025):

- **Thermal Output:** The Geretsried site is designed to produce up to 65 MW_{th} / 8 MW_e, demonstrating the scalability of the solution.
- **Drilling Depth:** The system includes wells drilled to a vertical depth of approximately 4,500 meters, where temperatures range between 150–165 °C.
- **Fluid Circulation:** Utilizing thermosiphon principles, the system enables efficient fluid circulation without requiring continuous pumping.

Numerous simulation studies have explored various closed loop geothermal system (CLGS) configurations (Chen et al. 2023; Beckeres et al. 2022; Chappidi et al. 2024; Qiao et al. 2024; Maaref et al. 2024; Nassan et al. 2024, 2025). However, most existing research relies on artificial models, highlighting the need for real-world applications.

Several pilot projects of CLGS have been implemented worldwide, including in Switzerland, Canada, and, most recently, the USA, where NOV is actively developing a project as of 2025 (Toews and Holmes 2021; Eavor Loop 2021; Gola et al. 2022; Kohl et al. 2002; Caines 2025).

The main drawback of CLS is slow heat transfer between the wellbore and the surrounding rock and this is due to the low thermal conductivity of the natural rocks (Wang et al. 2021). To overcome this problem, multi-wellbores working simultaneously with low flow rates will sustain these systems for long time scale.

This study presents a numerical simulation of a deep, multilateral, closed-loop well system, which is similar to the first project of stacked closed-loop wells in Geretsried, Germany. Prior to this, simulations of a single CLS in vertical and horizontal configurations are performed to demonstrate the superiority of multilateral CLS.

Methodology

The numerical model incorporates heat transfer processes within the wellbore and the surrounding geological formations, considering varying reservoir temperatures, thermal conductivities, and flow rates. Twelve (12) multilateral pairs were simulated in this study. Water contacts the wellbore wall directly in the horizontal sections to avoid reduction in heat conduction rate from the rock to the wellbore.

The simulations were conducted using the commercial software COMSOL Multiphysics (COMSOL 2022), which is based on the finite element method. Our previous work has successfully employed COMSOL for modeling geothermal doublets and fluid flow in porous media, demonstrating its capability to handle large-scale, coupled multiphysics (and occasionally multiphase) problems with reasonable computational efficiency (Nassan and Amro 2019, 2021; Nassan 2018, 2024). To optimize CPU time and streamline post-processing—particularly when using standard desktop computers—simplifying the simulated domain is essential.

Closed-loop system mathematical model

Two coupled physics modules were used in this study, namely: fluid flow and heat transfer in pipes, and heat transfer in solids. The governing mathematical models for both physics are listed below.

Wellbore mathematical model:

Mass conservation:

$$\frac{\partial(A\rho_f)}{\partial t} + \nabla_t \cdot (A\rho_f u e_t) = 0$$

Momentum conservation:

$$\begin{aligned} \rho_f \frac{\partial u}{\partial t} &= -\nabla_t p e_t - \frac{1}{2} f \frac{\rho_f}{d} \frac{d}{d_h} u + \mathbf{F} e_t \\ A &= \frac{\pi}{4} d_{pipe}^2 \\ d_h &= d_{pipe} \\ Re &= \frac{\rho_f d_h u}{\mu} \\ C_A &= \left[-2.457 \ln \left(\left(\frac{7}{Re} \right)^{0.9} + 0.27 \left(\frac{0.26}{d_h} \right) \right) \right]^{16} \\ C_B &= \left(\frac{37530}{Re} \right)^{16} \\ f_d &= 8 \left[\left(\frac{8}{Re} \right)^{12} + (C_A + C_B)^{-1.5} \right]^{\frac{1}{12}} \end{aligned}$$

Energy conservation:

$$(hZ)_{eff} = \frac{1}{\frac{1}{Z_0 h_{int}} + \frac{1}{Z_0 h_{ext}} + \int \left(\ln \left(\frac{r_n}{r_n - 1} \right) \right) \frac{1}{2\pi k_{wall,n}} dr}$$

Fig. 5 illustrates how the wall layers are handled in Comsol Multiphysics.

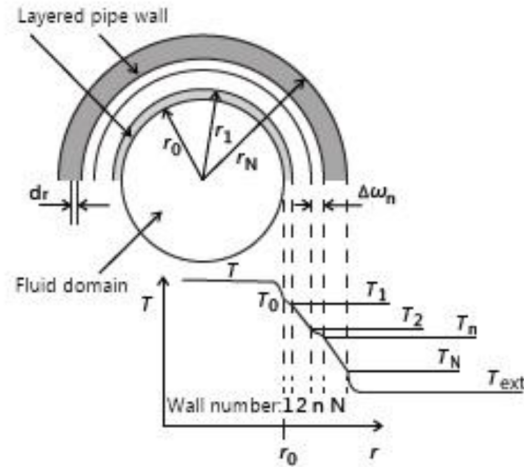


Figure 5—Wellbore wall layers' handling method in Comsol Multiphysics (Comsol-pipe flow module 2022).

$$Nu_{lam} = 3.66 \text{ for round pipes}$$

$$Nu_{turb} = \frac{\left(\frac{f_d}{8}\right)(Re-1000)Pr}{1+12.7\sqrt{\frac{f_d}{8}}\left(Pr^{\frac{2}{3}}-1\right)}$$

$$Pr = \frac{C_{pw}\mu}{k}$$

$$Nu = \max(Nu_{lam}, Nu_{turb})$$

$$Z = \pi d_{pipe}$$

$$h_{int} = Nu \frac{k}{d_h}$$

Rock mathematical model:

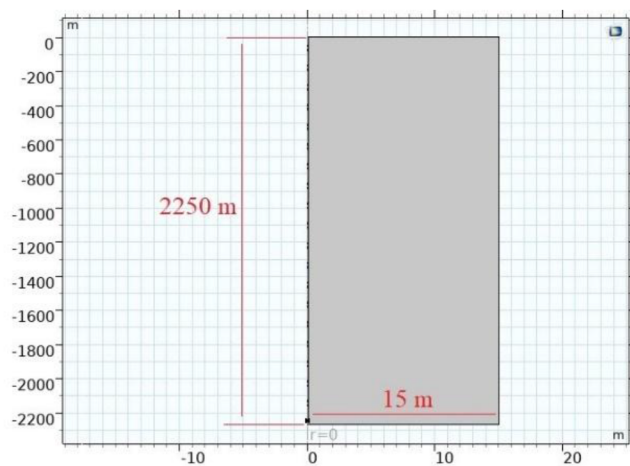
Energy conservation:

$$\rho C \frac{\partial T_2}{\partial t} + \rho C_p u \nabla T_2 + \nabla \cdot (-k \nabla T_2) = Q$$

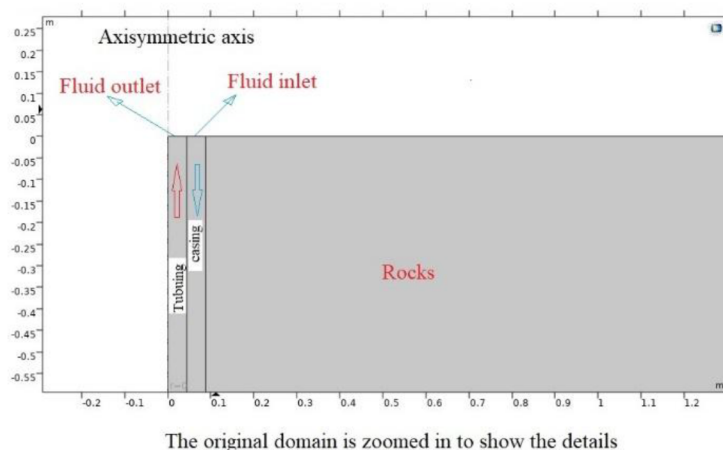
Nomenclature is explained in the symbols section.

Simulated models

In this study, three models are compared: the vertical coaxial single-loop, the horizontal single-loop, and the multilateral-loop configuration with 12 lateral pairs. The three models are illustrated in Figs. 6 through 8.



(a) Vertical single-loop model



(b) Zoomed in model of (a) to show the inlet and outlet

Figure 6—Axisymmetric vertical coaxial single-loop model (after Nassan et al. 2025).

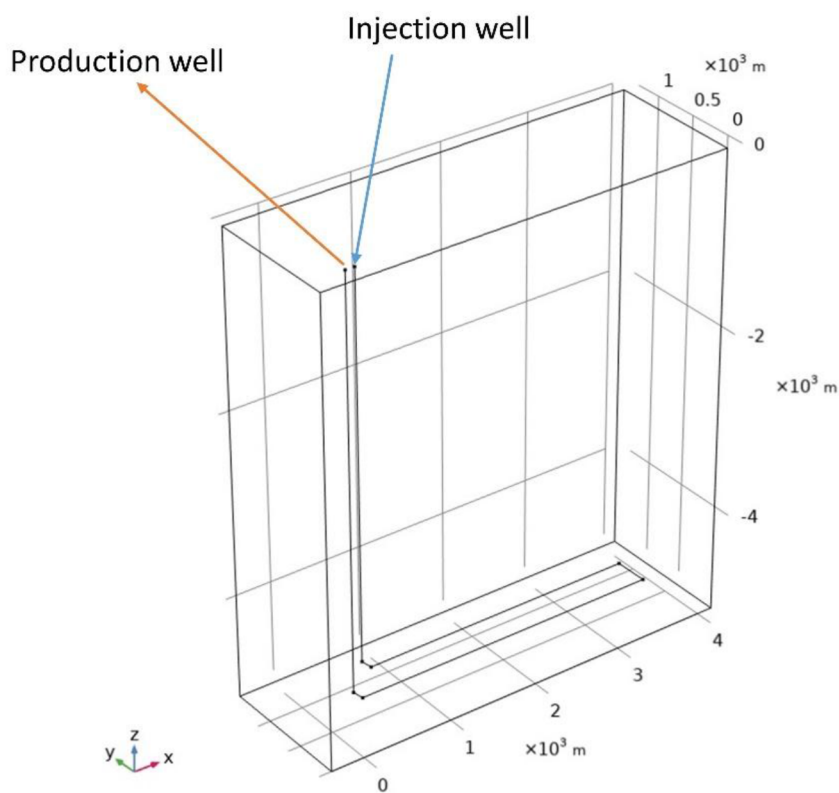


Figure 7—Horizontal single-loop model.

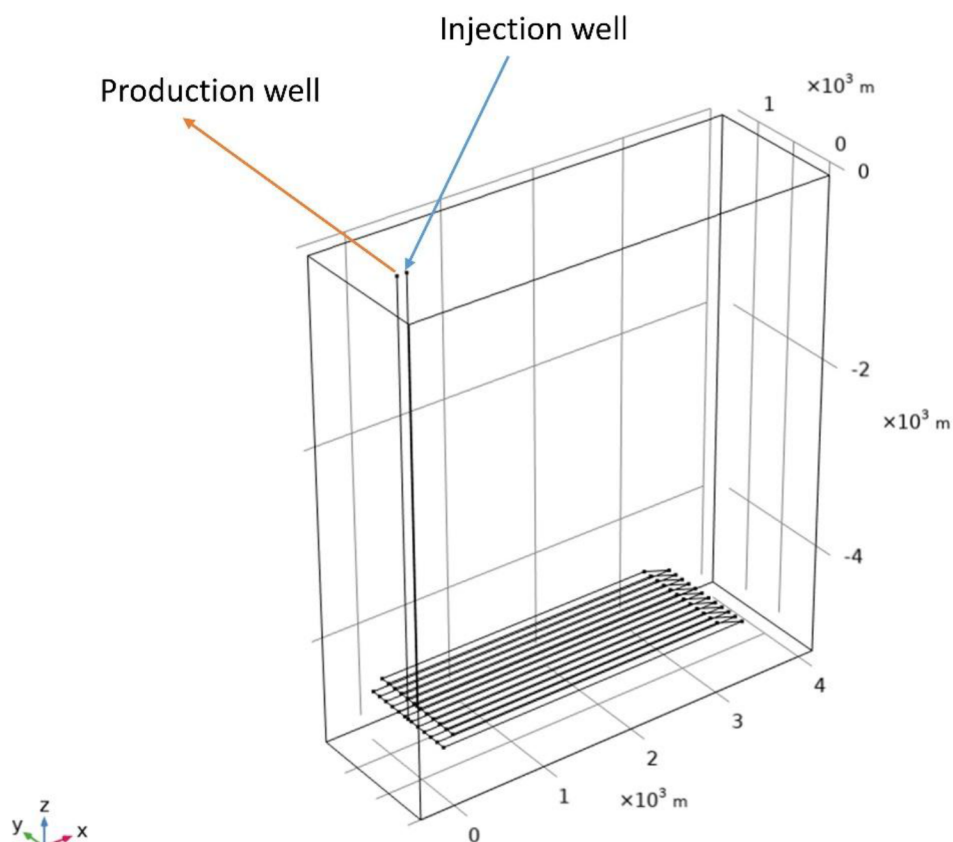


Figure 8—Multilateral CLS with 12 horizontal pairs.

Vertical single-loop model. The first model is a single vertical coaxial loop model that have been simulated by Nassan et al. (2025). The model represents a real wellbore model drilled and abandoned in Germany. Nassan et al. (2025) simulated the efficiency of a single-loop with the data of the model listed in table 1. Several rates were considered: 0.5 kg/s, 1 kg/s, 2 kg/s, 4 kg/s, 10 kg/s, 20 kg/s. Fig. 6(a) illustrates single coaxial loop in axisymmetric mode. Fig. 6(b) is a zoomed in sketch of the top left corner of the domain in 6(a) to show the inlet and outlet of the system.

Table 1—Parameters of the vertical coaxial single-loop, horizontal single-loop, and multilateral-loop models.

Parameter, unit	Value/vertical coaxial single-loop configuration	Value/ horizontal single-loop configuration	Value/multilateral-loop configuration
Rock thermal conductivity, W/m. K	2	2	2
Water thermal conductivity, W/m. K	0.6	0.6	0.6
Rock heat capacity, J/kg. K	900	900	900
Water heat capacity, J/kg. K	4200	4200	4200
Rock density, kg/m ³	2250	2250	2250
Water density, kg/m ³	1000	1000	1000
Vertical depth (z), m	2250	4500	4500
Horizontal length, m	-	2×3300	12×2×3300
Casing diameter, inch	9 5/8 (1585m), 7 (2249 m)	6	6
Tubing diameter, inch	3 1/2	-	-
Bottom heat flux, W/m ²	0.065	0.120	0.120
Injection temperature, °C	10	50	50

Parameter, unit	Value/vertical coaxial single-loop configuration	Value/ horizontal single-loop configuration	Value/multilateral-loop configuration
Thermal gradient, °C/m	0.063	0.033	0.033
Injection rate, kg/s	0.5, 1, 2, 4, 10, 20	25	85, 120
Simulation time	360 days	30 a	100 a
Initial pressure, Pa	$-z \times 1000 \times 9.81$	$-z \times 1000 \times 9.81$	$-z \times 1000 \times 9.81$
Initial temperature, °C	$10 + 0.063 \times z$	$16.5 + 0.033 \times z$	$16.5 + 0.033 \times z$

Horizontal single-loop model. The second model is a single horizontal loop as depicted in Fig. 7. The main parameters of the model are listed in table 1.

Multilateral-loop model. The third model is a multilateral horizontal loop as depicted in Fig. 8. The main parameters of the model are also listed in table 1.

Model validation

The reliability of the simulation using Comsol Multiphysics is validated by simulating the vertical CLS by CMG-STARs software. A model was created by CMG builder and the new added feature of CLS to CMG was used and a case was simulated and in a previous publication a validation study was shown (Nassan et al. 2025). The results showed good agreement between both software.

Results and discussion

In the following we will report and discuss the results of the three simulated cases.

Vertical single-loop model

The vertical CLS results shown in this study are a simulated case study conducted for a high geothermal gradient well exists in Germany. The well is drilled to a depth of 2300 m. Fig. 9 shows the temperature at several flow rates while Fig. 10 depicts thermal energy extracted accordingly. It can be seen that, the lower the flow rate, the higher the temperature while the extracted energy behaves vis versa. A comparison among all flow rates scenarios shown in Fig. 11 shows that the best case scenario is at low flow rate (less than 4 kg/s).

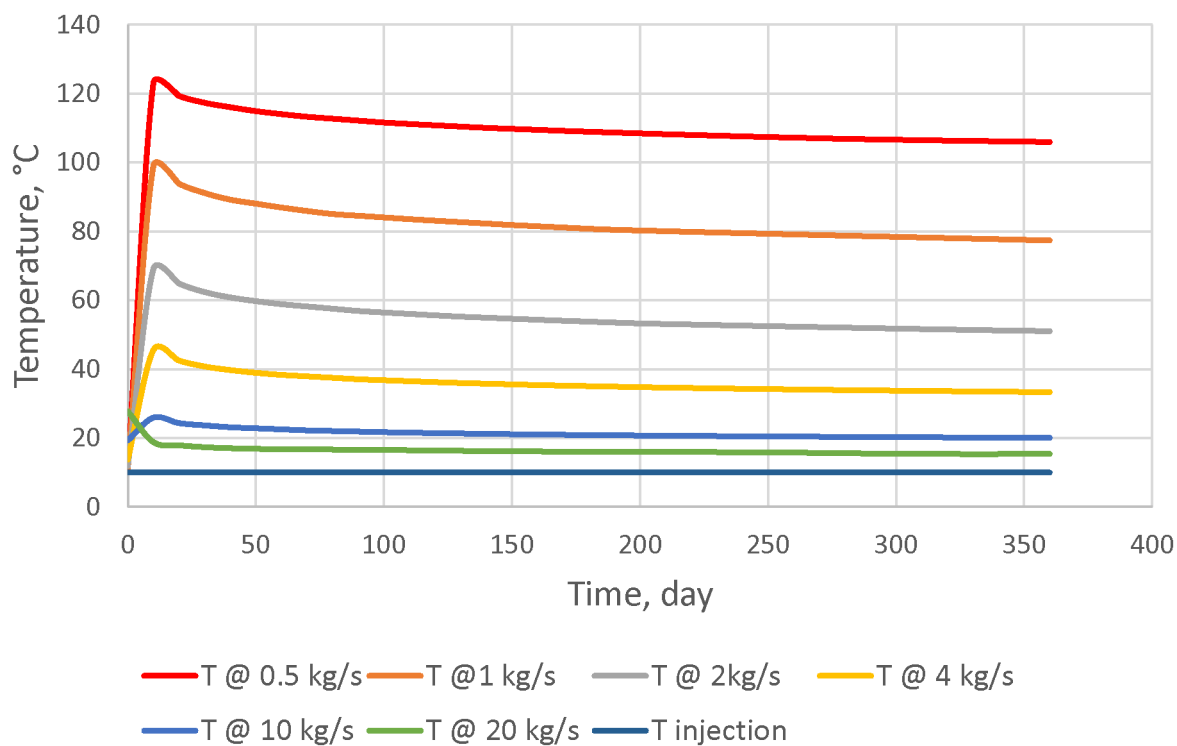


Figure 9—Produced water temperature at different rates for vertical single CLS (after Nassan et al. 2025).

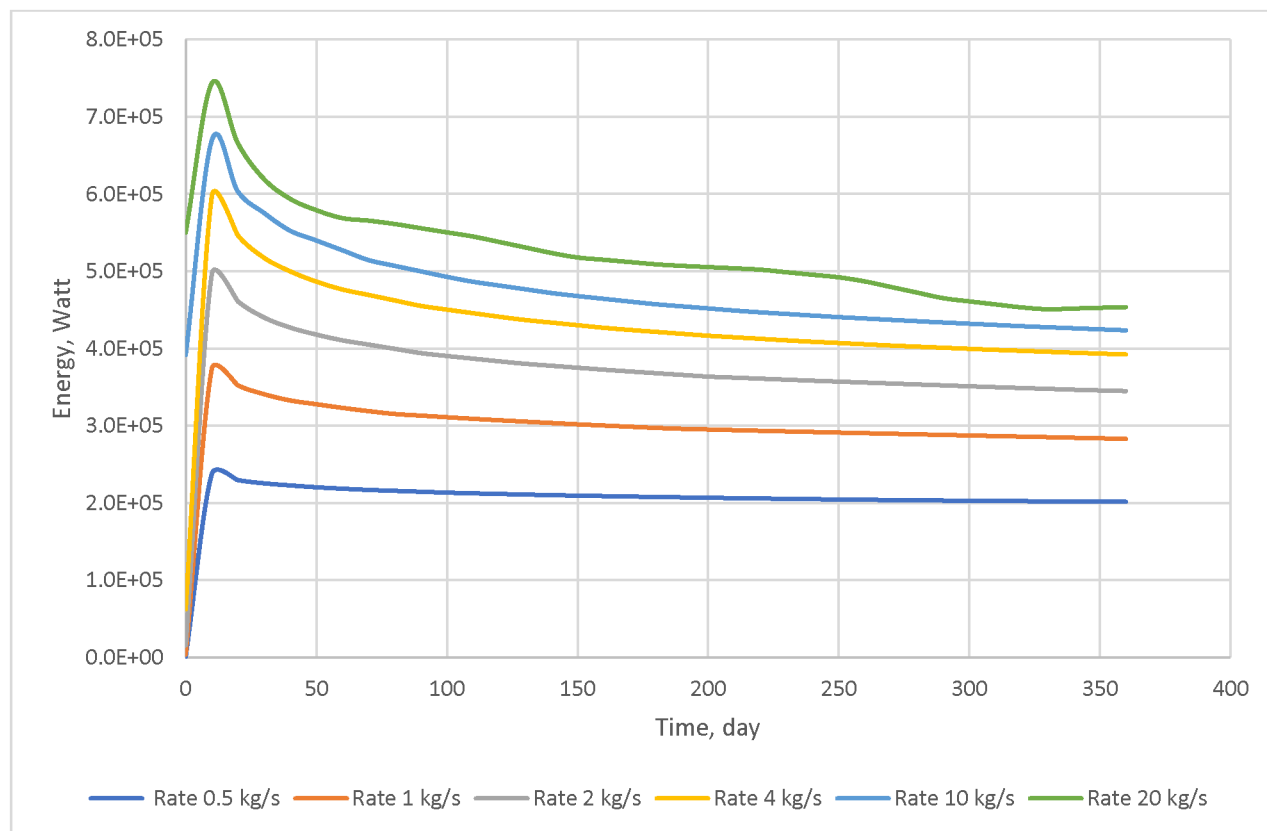


Figure 10—Extracted energy at different flow rates for vertical single CLS based on $\Delta T = T - T_{\text{injection}}$ (after Nassan et al. 2025).

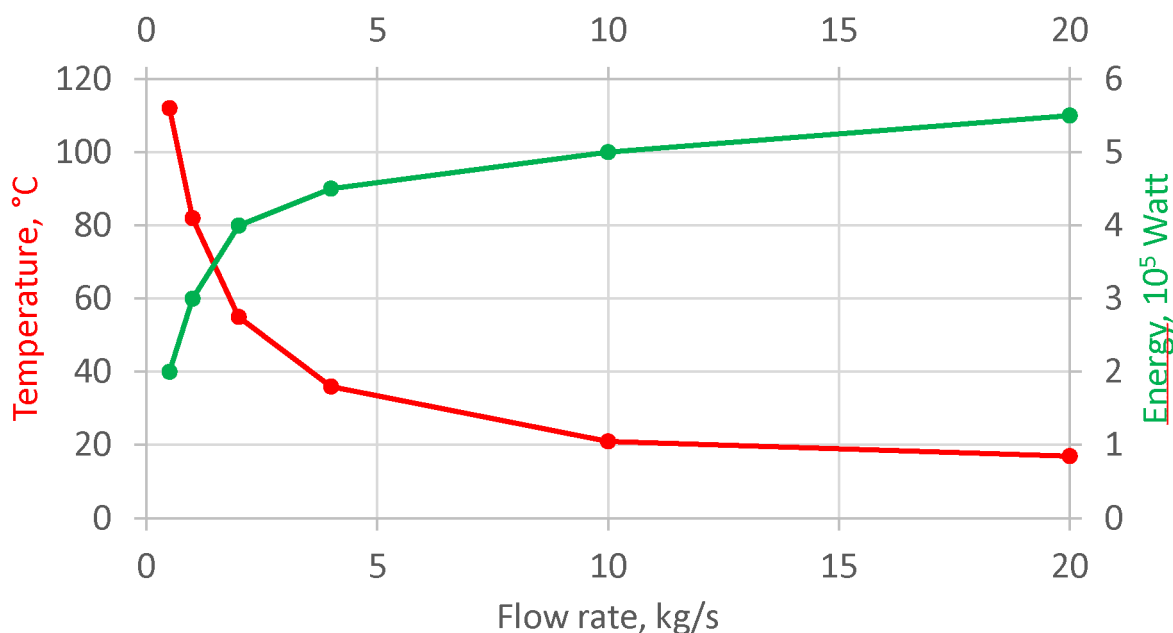


Figure 11—Comparison of all flow rates scenarios for vertical single CLS.

Horizontal single-loop model

One horizontal loop was simulated at a rate of 25 kg/s. The results are depicted in Fig. 12. It can be seen that the temperature drops below boiling point of water after only five years which indicates that the system cannot be used for electricity generation even at this low flow rate. Of course decreasing the flow rate will raise the temperature of the produced fluid but will not be efficient for electricity generation.

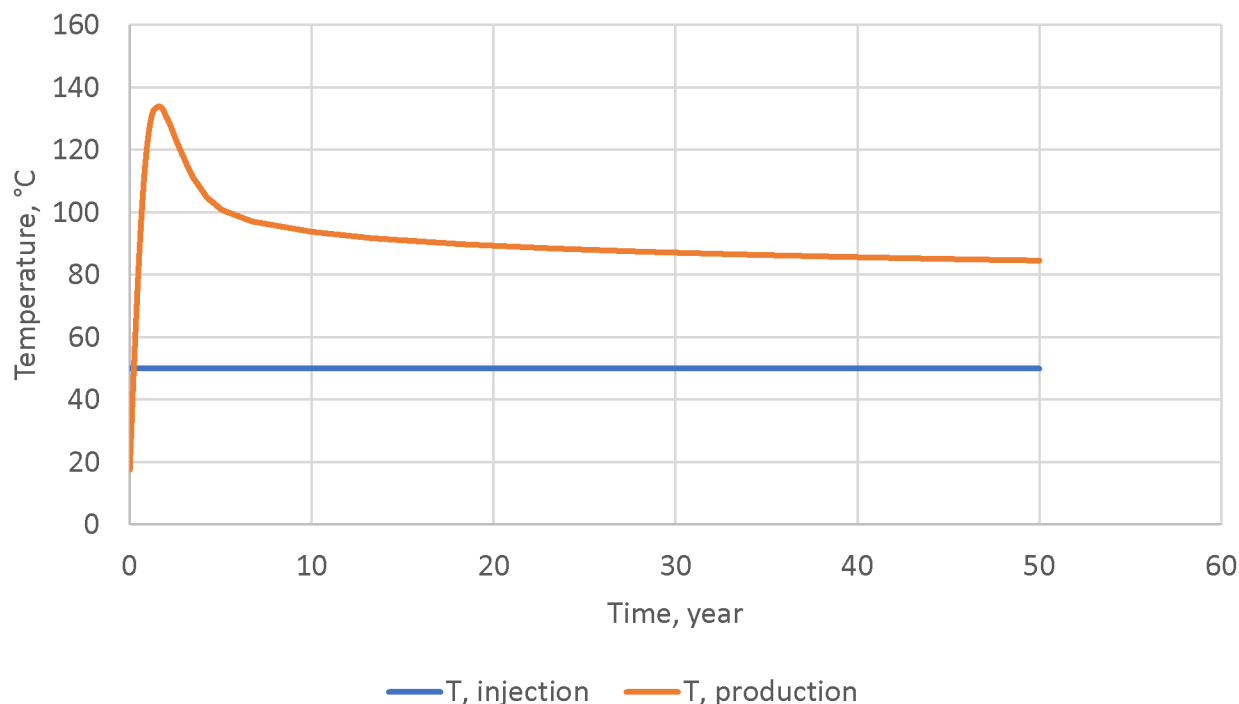


Figure 12—Temperature of the produced fluid in the single horizontal-loop configuration.

Multilateral-loop model

Multilateral model encompasses 12 horizontal pairs. Two scenarios were simulated with real rock thermal conductivity from south German drilling location at Geretsried. The same flow rate 85 kg/s was tested but

with 100 m and 200 m offset between horizontal sections. From Fig. 13, it can be seen that the effect of increasing the spacing from 100 m to 200 m is relatively little. However, if implemented, it would be better especially if the system is intended for more than 50 years. Fig. 14 shows the 3D domain at four different time steps.

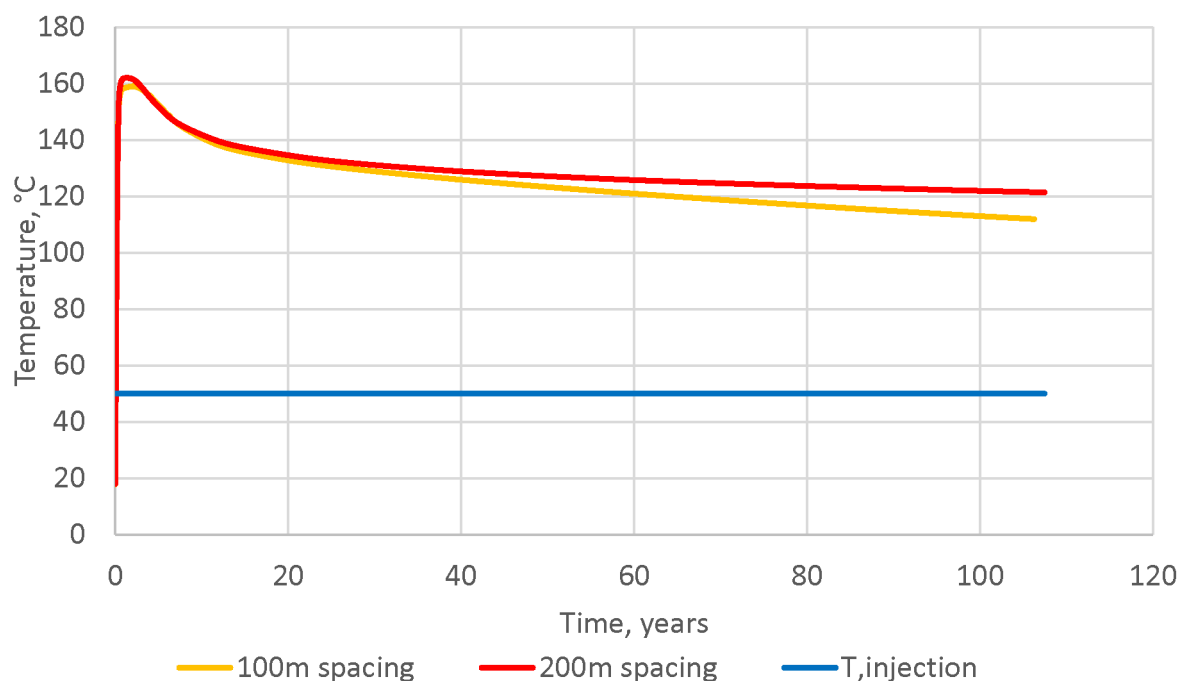


Figure 13—Comparison of the produced fluid temperature in the multilateral-loop model (12 pairs) with 100 m and 200 m spacing between horizontal laterals, at a flow rate of 85 kg/s.

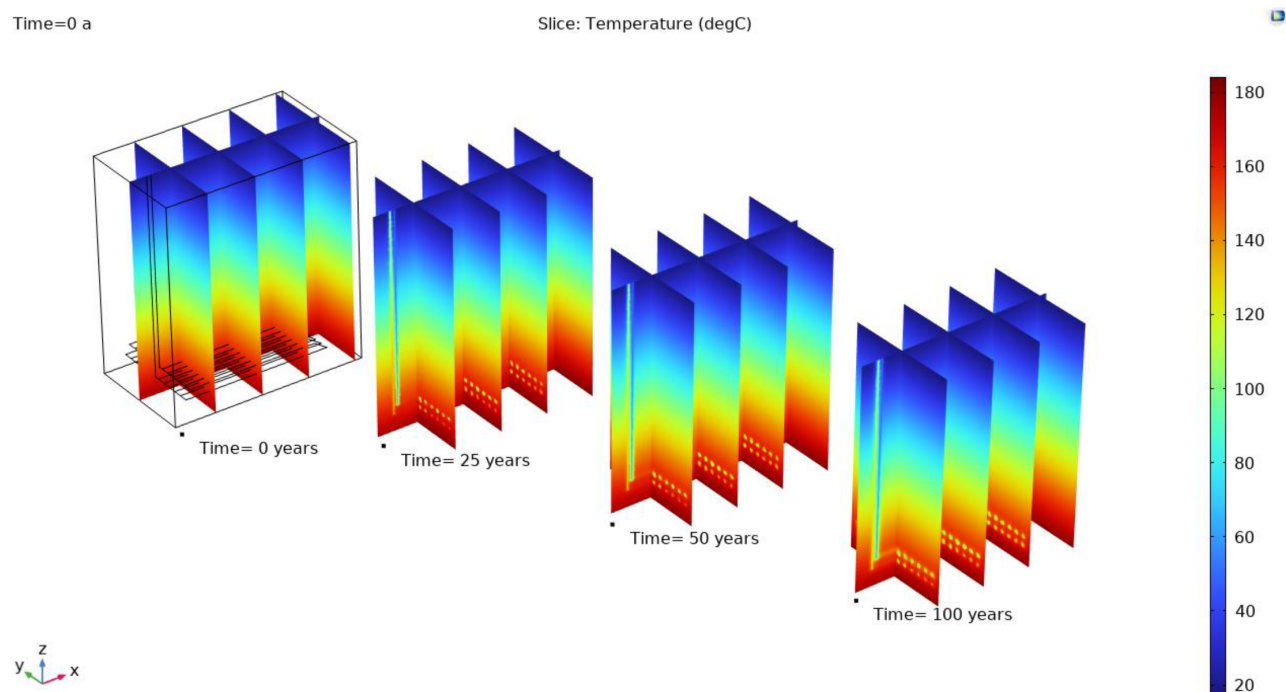


Figure 14—Temperature distribution around the wellbores with 200 m spacing, shown at different time steps in 3D using selected slices for improved visualization.

It should be noted that the model is built without any kind of insulation between the casing and the formation (referred to direct contact). This interpret the gradual increase of temperature of the produced fluid in the first few months until the region around wellbore being heated enough, so the losses after that will be almost negligible.

If insulation for the production well is considered using three different thermal conductivities (0.8, 0.5, and 0.15 W/m. K) for a cement sheath with a thickness of 2 cm, the results shown in Fig. 15 are obtained. It is evident that the insulating effect of the cement sheath has a minimal impact on increasing the extracted heat. This can be attributed to the high flow rate of the produced fluid and the relatively low thermal conductivity of the surrounding rock (2 W/m·K).

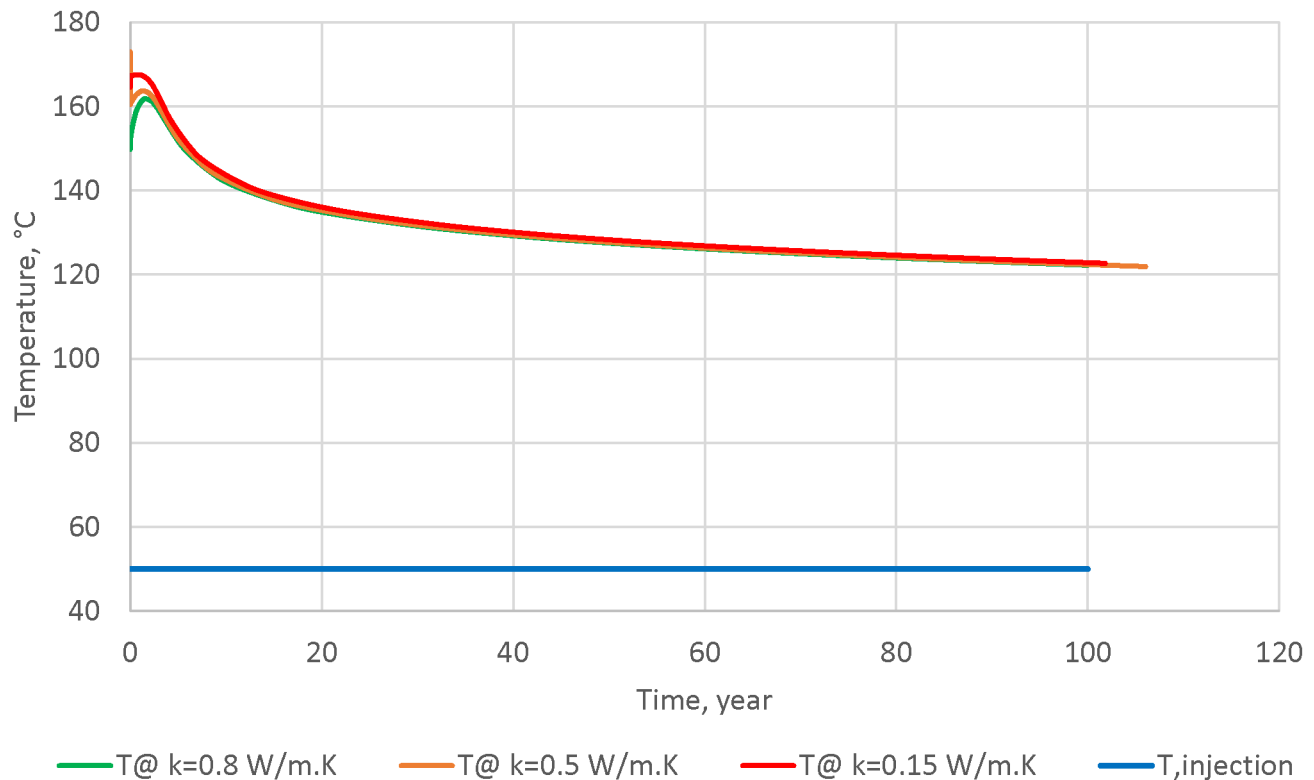


Figure 15—Comparison of produced fluid temperature in the multilateral-loop model (12 pairs) at three different cement sheath thermal conductivities, with a flow rate of 85 kg/s.

The final tested scenario is an abnormal geothermal gradient of 6.3 °C/100m. Considering two flow rates at 85 and 120 kg/s, the results are shown in Fig. 16. It can be seen that the geothermal gradient is extremely important. At 85 kg/s, a temperature of more than 200 °C is guaranteed for over 100 years and for the first 10 years above 250 °C. Such system is much better than the original scenario at 3.3 °C/100 m geothermal gradient (similar to the Geretsried system in Germany).

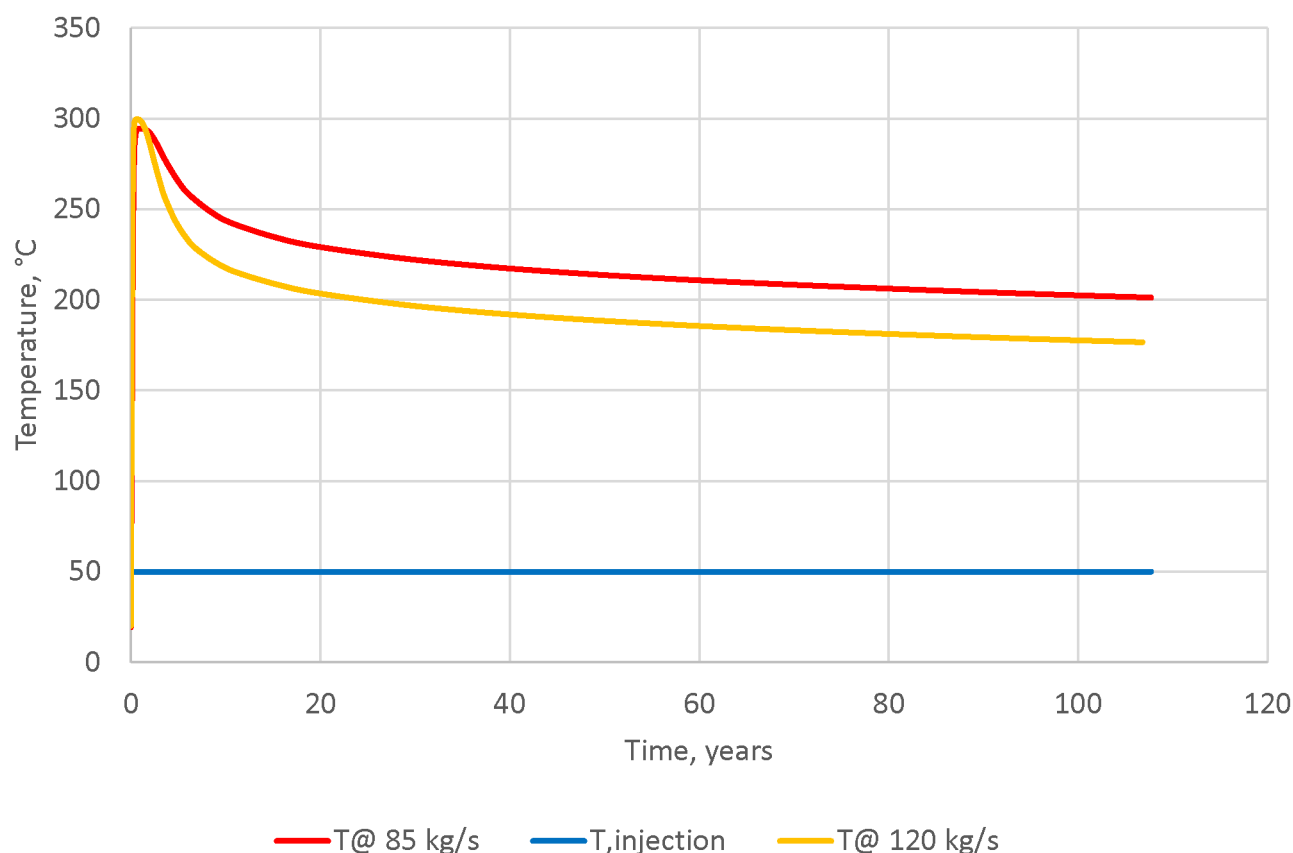


Figure 16—Comparison of the produced fluid temperature in the multilateral-loop model (12 pairs) at a geothermal gradient of 6.3 °C/100 m, for flow rates of 85 and 120 kg/s.

Discussion

As shown in the results above, the multilateral configuration provides a higher and more stable temperature over a period of more than 40 years, and can produce fluid at a rate of 85 kg/s—suitable for powering a steam turbine. In contrast, the single horizontal loop configuration delivers fluid at a significantly lower temperature and a reduced flow rate of 25 kg/s.

The vertical coaxial CLS appears to be inefficient at higher flow rates (above 4 kg/s) and is only suitable for space heating applications with a heat pump. This limitation is due to the low heat extraction rate, despite the high geothermal gradient used in the simulated case study (6.3 °C/100 m).

There are two main concerns with applying CLS in vertical wells due to the low outlet fluid temperature:

1. Tubing insulation – This challenge can be mitigated by using dual vacuum pipe technology, which minimizes heat loss from the produced fluid (Maver, 2024).
2. Heat losses in transport pipelines – The only effective solution is to use wells near populated areas to reduce transport distances and associated losses.

Moreover, such systems are only economically viable if the well has already been drilled. Drilling a new well solely for CLS application is not cost-effective due to the low heat extraction rate.

Deep CLS with a multilateral configuration offers a novel and sustainable approach to harnessing geothermal energy, avoiding the typical hazards associated with traditional geothermal systems—such as corrosion, induced seismicity, and low injectivity. However, challenges like high capital costs and technical complexities currently limit the large-scale deployment of this technology. At present, it remains under active development.

The ongoing project in Geretsried, Germany, along with two additional planned projects in other regions of the country, will provide valuable data upon completion. The outcomes of these projects will help guide future advancements in CLS technology. Moreover, testing and developing new tools as part of these initiatives may lead to cost reductions, paving the way for broader global adoption.

Lastly, while multilateral CLS systems can be implemented in almost any region, the results indicate that heat extraction is significantly more efficient in areas with higher geothermal gradients.

Conclusions and outlook

The global energy landscape is undergoing a significant transition toward CO₂-free fuels and energy sources. International policy frameworks and climate reports consistently recommend that industrialized nations, such as Germany, achieve full decarbonization by mid-century, while less developed countries are expected to reach this target by the end of the century. Despite these goals, the 2023 global energy consumption report indicates that fossil fuels continue to dominate as the primary energy source worldwide.

To enable a successful transition to renewable energy, the deployment of novel technologies is essential, as conventional renewable sources—such as wind and solar—are geographically constrained. Geothermal energy presents a promising alternative: it is a globally available, baseload-capable resource that can be exploited continuously, independent of weather or seasonal variability. However, its broader utilization demands technological innovation. In recent years, numerous start-ups and patented technologies have emerged, many of which show substantial promise in addressing existing limitations.

This study evaluates the performance of three closed-loop geothermal configurations through numerical modeling. The results indicate that vertical closed-loop systems are only viable at low flow rates, particularly when existing wells are repurposed and located near populated areas. Similarly, single horizontal loop systems operate at lower flow rates and exhibit limited thermal output compared to multilateral configurations. Among the analyzed configurations, the multilateral closed-loop system demonstrates superior long-term operational stability and thermal performance. Nonetheless, economic feasibility remains a critical constraint.

While the first phase of the Stacked CLS project in Geretsried, Germany, will become operational soon, and two additional projects are planned in Hanover and Ulm, high capital expenditure continues to hinder the widespread adoption of this technology. Further research and technological advancement are necessary to reduce costs and improve scalability, thereby enabling closed-loop geothermal systems to contribute meaningfully to the global energy transition.

Symbols

Symbol	Description	Unit
A	Pipe cross-section area	m^2
e_t	Tangential vector	-
ρ_f	Fluid density	kg/m^3
ρ	Rock density	kg/m^3
t	Time	s
u	Fluid velocity	m/s
p	Fluid pressure	Pa
f_d	Friction factor	-
F	Volume force term	N/m^3
d_h	Hydraulic diameter	m
Re	Reynolds number	-
$C_{p,w}$	Fluid heat capacity (water)	$\text{J}/(\text{kg} \cdot ^\circ\text{C})$
C_p	Rock heat capacity	$\text{J}/(\text{kg} \cdot ^\circ\text{C})$
T	Fluid temperature	$^\circ\text{C}$
T_2	Rock temperature	$^\circ\text{C}$
k	Thermal conductivity (solid/rock)	$\text{W}/\text{m} \cdot ^\circ\text{C}$
Nu	Nusselt number	-
Pr	Prantl number	-
Q_{wall}	Heat exchange through the pipe wall	W/m
h	Effective heat transfer coefficient	$\text{W}/(\text{m}^2 \cdot ^\circ\text{C})$
Z	Perimeter of the pipe	m

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